

MINGRUI XU

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EDUCATION

Carnegie Mellon University	Entertainment Technology	M.E.T.	Aug. 2025 - Present
University of Utah	Games	B.S.	Aug. 2021 - May 2025

- GPA: 3.778/4.0
 - Honors: Dean's List (Aug. 2021, May 2022, Dec. 2022, May 2023, and Dec. 2024); Part-Time Dean's List (Dec. 2023)
- University of Massachusetts Boston | Non-Degree Undergraduate Program (credits transferred to the University of Utah)
Aug. 2020 - Jul. 2021

CAPSTONE & ACADEMIC PROJECTS

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| Beam Down (Third-Person Top-Down Boss Rush Game) | Game Designer & Technical Artist |
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- Aug. 2024 - May 2025
- Led the end-to-end design of the player upgrade system, including skill mechanics, upgrade UI interactions, skill naming, and narrative descriptions, ensuring close integration between gameplay and story.
 - Collaborated on the combat mechanics and visual design of enemies and the boss in the first world, working with artists and programmers to implement the content in the game.
 - Collaborated on enemy abilities and visual effects (VFX), improving combat readability, player feedback, and immersion.
 - Contributed to the level design of the first world to support both gameplay and narrative requirements.
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| Building Virtual Worlds | Game Designer, Technical Artist & Sound Designer | Aug. 2025 - Dec. 2025 |
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- Worked in rotating teams over a 12-week period to complete four short game-development projects lasting one, two, or three weeks. Served as a game designer by proposing initial gameplay concepts and iterating on them based on player feedback.
 - Created sound effects and music to support each game's atmosphere and communicate gameplay information effectively.
 - Assisted with VFX design to provide clear and effective player feedback.

INTERNSHIP EXPERIENCE

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| Beijing Tianshen Interactive Technology Co., Ltd. | VFX Designer & Live Operations Designer |
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- Apr. 2024 - Jun. 2024
- VFX Design Responsibilities:**
- Collaborated across game design and programming teams, using Unreal Engine 5 (UE5) and Unity to design and implement core in-game visual effects.
 - Created VFX concepts for magic, explosions, fire, water, and environmental ambience based on the project's visual style and gameplay requirements, ensuring consistency with the game world.
 - Produced VFX animation sequences in UE5 and refined effect parameters in Adobe After Effects (AE), precisely controlling timing, duration, and fade-in/fade-out behavior for combat and interactive scenes.
 - Coordinated with the art, design, and audio teams to align VFX with the overall visual style and sound rhythm, strengthening audiovisual impact and the player experience.
- Live Operations Responsibilities:**
- Managed the player-feedback loop for the Thai market by monitoring bug reports, coordinating fixes, and supporting regular version updates to maintain a smooth experience for overseas users.
 - Analyzed Thai player behavior and corrected localization and cultural-adaptation issues, including translation discrepancies, cultural sensitivities, and aesthetic preferences, to improve local acceptance.
 - Optimized VFX and UI design by adjusting interface layouts to Thai reading habits and adding locally appropriate interactive elements, such as tutorials and feedback pop-ups, to improve usability.

- Improved user-feedback channels, regularly collected and analyzed player suggestions, and produced optimization reports to support version iteration and continuous gameplay improvements.

RESEARCH EXPERIENCE

Applications of Psychological Principles in Game Design May 2023 - Jul. 2023

- Project's Faculty Mentor is Professor Geoff Kaufman, Human-Computer Interaction Institute, Carnegie Mellon University
- Studied core principles of game design and explored ways to integrate modern human-computer interaction with game development.
- Applied psychological concepts such as flow, immersion, and persuasion to game design, examining how challenge, novelty, and uncertainty can improve player engagement and retention.
- Participated in seminars on the effects of games on human cognition and behavior; completed and presented a player-behavior analysis of FPS titles including VALORANT, Apex Legends, and Counter-Strike: Global Offensive.

PUBLICATION

Mingrui Xu. "An Analysis of How First-Person Shooter Games Retain Players of Different Skill Levels."

[C] Proceedings of ICSPHS 2024, ICSPHS_0667. DOI: 10.54254/2753-7048/39/20240667

SKILLS & ADDITIONAL INFORMATION

- Programming Languages: Python, Java, C#
- Game Engines: Unreal Engine (primarily UE5; VFX and level development), Unity (gameplay prototyping and interaction implementation)
- Design & Production Tools: Adobe After Effects (VFX post-production), Adobe Premiere Pro (video editing), Maya (3D modeling and animation), Photoshop (UI and icon design), Houdini (advanced VFX simulation), Adobe Illustrator (vector graphics)
- Writing: Experienced in worldbuilding, long- and short-form fiction, and scripted content for Vtubers.